Assignment 6 - with responses

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Inhalte

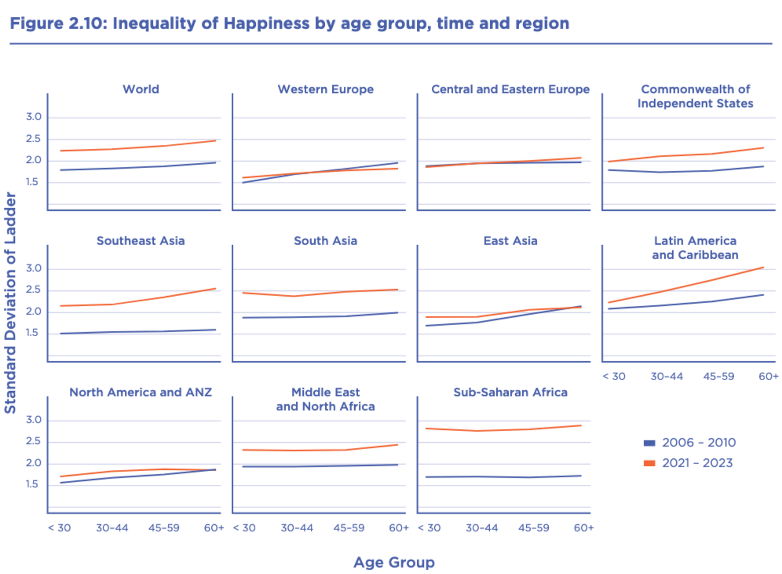


Figure 1: Happiness in different parts of the world over time.

Introduction

Dataset:

A subset of the dataset form the 2024 World Happiness Report, which includes the countries for which 15 datapoints or more were available. The report presents the results of a yearly survey of people's happiness in different parts of the world. The initiative is a collaboration of Gallup, the Oxford Wellbeing Research Centre, the UN Sustainable Development Solutions Network, and the WHR's Editorial Board.

The central results are in the first chapters of the report, summarised in its Figure 2.1.

In general, the initiative seeks to highlight happiness as a central indicator of societal progress; that it can be measured; and that its measurement can form an empirical basis for interventions to promote happiness.

In each survey round, around 1000 people from each country are asked several questions related to happiness. In one question they are asked to imagine a ladder with steps from 0 to 10, and to rate their overall life happiness on it, with higher steps representing higher happiness ("Cantril ladder", corresponding to the variable `Life.Ladder` in the dataset). Further questions refer to positive and negative emotions experienced the past day; and to possible causes of happiness like GDP, social support, freedom to make life choices, and perceived corruption. Below you see the first rows of the dataset.

```
d <- read.csv("countries_n15over.csv")
head(d, 4)
```

```
##   Country.name year Life.Ladder Log.GDP.per.capita Social.support
## 1  Afghanistan 2008    3.723590           7.350416      0.4506623
## 2  Afghanistan 2009    4.401778           7.508646      0.5523084
## 3  Afghanistan 2010    4.758381           7.613900      0.5390752
## 4  Afghanistan 2011    3.831719           7.581259      0.5211036
##   Healthy.life.expectancy.at.birth Freedom.to.make.life.choices Generosity
## 1                             50.5                    0.7181143  0.1640551
## 2                             50.8                    0.6788964  0.1872966
## 3                             51.1                    0.6001272  0.1178605
## 4                             51.4                    0.4959014  0.1600984
##   Perceptions.of.corruption Positive.affect Negative.affect
## 1                 0.8816863       0.4142970       0.2581955
## 2                 0.8500354       0.4814214       0.2370924
## 3                 0.7067661       0.5169067       0.2753238
## 4                 0.7311085       0.4798347       0.2671747
```

Packages

```
library(kableExtra)  # tables
library(ggplot2)     # graphics
library(ggokabeito)  # colorblind friendly color palettes
library(brms)        # Bayesian modelling
library(performance) # icc
```

Part 1: Data wrangling

Exercise 1.

Choose 5 countries from a given continent, and create a new, smaller dataset with only the data from those countries (you can use the `subset()` function for this).

```
# The full dataset
d_all <- read.csv("countries_n15over.csv")

# The subset
d <- subset(d_all, (d_all$Country.name %in% "Portugal")|
              (d_all$Country.name %in% "Italy")|
              (d_all$Country.name %in% "Greece")|
              (d_all$Country.name %in% "France")|
              (d_all$Country.name %in% "Latvia"))
# For convenience:
```

```
names(d) <- tolower(names(d)) # lowercase variable names
d$year <- as.numeric(d$year)  # numeric variable type
```

Exercise 2.

Formulate a hypothesis to test with your dataset. The hypothesis should be testable using `Life.Ladder` as criterion and one or more of the other variables as predictors.

Happiness (as measured in 'Life.Ladder') increases with social support ('Social.support'), although this relationship varies between countries (and over time).

Exercise 3.

Illustrate the pattern in the data that is relevant to the hypothesis with a graph. Design the graph in such a way, that (a) two (or more) levels of aggregation are visible, and (b) not only information about means, but also information about the variance around those means is visible, (c) the figure axes have comprehensible labels.

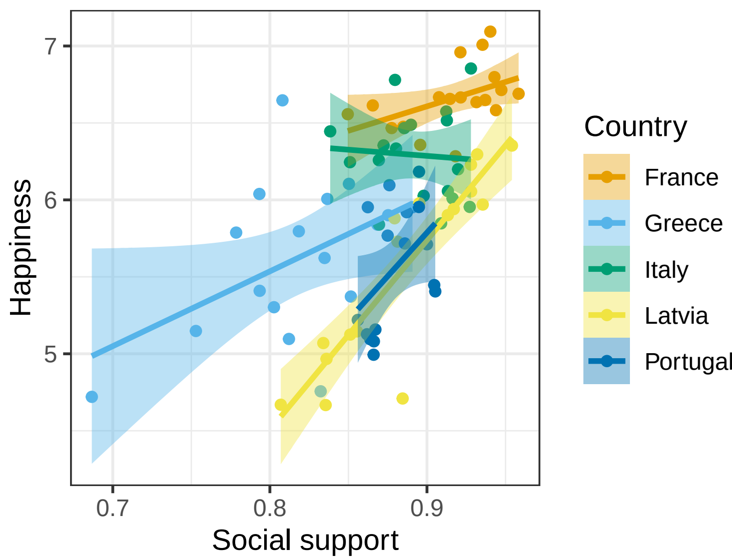


Figure 2: Happiness as a function of social support in several European countries.

Part 2: Data analysis

Exercise 4.

Specify and test a multilevel model to test your hypothesis. Let your model contain at least two levels of aggregation, and include parameters to assess not only means ("fixed effects") but also variability around those means ("random effects"). Test it both by computing posterior parameter values, and by comparing alternative models (with vs. without a parameter of interest) using Bayes factors.

The model below assesses the extent to which social support can predict happiness, and inhowfar this varies between countries. The hypothesis is tested not only by estimating the parameters of one model, but also by comparing models with vs. without a parameter of interest using Bayes factors.

```
# One level
set.seed(9876)

output <- capture.output(
  m0 <- brm(life.ladder ~ social.support
```

```r
           , data = d
           , sample_prior = TRUE
           , warmup = 1000
           #, iter = 11000
             , chains = 4
             , cores = 4
           , save_pars = save_pars(all = TRUE)
           )
)
# Two levels with random intercept for country.name
output <- capture.output(
  m1 <- brm(life.ladder ~ social.support + (1|country.name)
           , data = d
           , sample_prior = TRUE
           , warmup = 1000
           #, iter = 11000
             , chains = 4
             , cores = 4
           , save_pars = save_pars(all = TRUE)
           )
)

# Two levels wth random intercept and random slope
# for country.name
output <- capture.output(
m2 <- brm(life.ladder ~ social.support + (social.support|country.name)
           , data = d
           , sample_prior = TRUE
           , warmup = 1000
           #, iter = 11000
             , chains = 4
             , cores = 4
           , save_pars = save_pars(all = TRUE)
           )
)

output <- capture.output(bf_10 <- bayes_factor(m1, m0))
output <- capture.output(bf_21 <- bayes_factor(m2, m1))
bf_10
```

```
## Estimated Bayes factor in favor of m1 over m0: 179096733.99324
```

```r
bf_21
```

```
## Estimated Bayes factor in favor of m2 over m1: 0.70855
```

```r
1/bf_21$bf
```

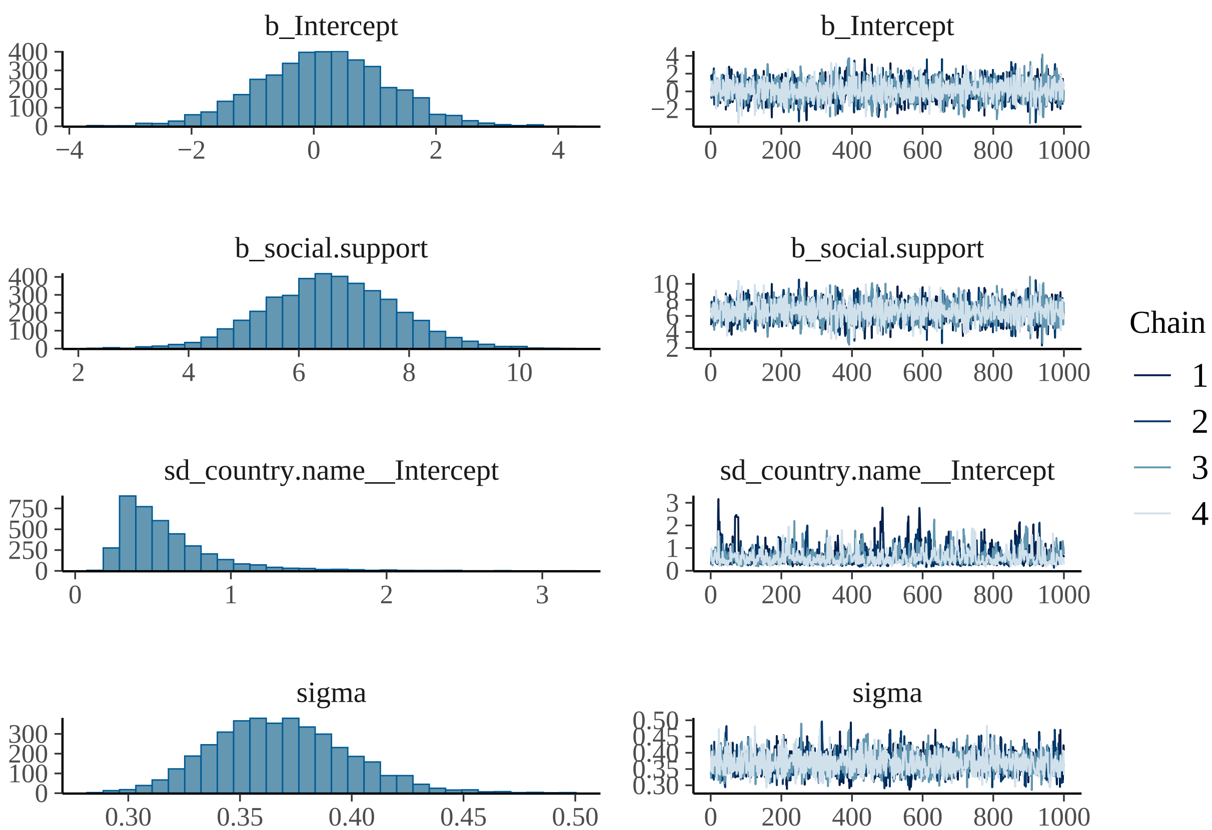
```
## [1] 1.411325
```

Model `m1` fairs better than model `m0` ($BF_{10} > 100$). This indicates there is evidence that happiness indeed varies between countries. In contrast, model `m2` could account for the data less well than model `m1` ($BF_{21} = 0.71$, $1/BF_{21} = BF_{12} = 1.41$). Hence, although the graph suggests the slopes may vary between countries, this variability did not seem large enough to justify the increased complexity of `m2`. Looking back at the graph, this may be explained by a restriction of range and/or a ceiling effect in both variables. Overall, the final model we shall use for further analysis is `m1`.

```r
summary(m1)
```

```
##  Family: gaussian
##   Links: mu = identity
## Formula: life.ladder ~ social.support + (1 | country.name)
##    Data: d (Number of observations: 86)
##   Draws: 4 chains, each with iter = 2000; warmup = 1000; thin = 1;
##          total post-warmup draws = 4000
##
## Multilevel Hyperparameters:
## ~country.name (Number of levels: 5)
##               Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## sd(Intercept)     0.59      0.34     0.24     1.52 1.00      918     1258
##
## Regression Coefficients:
##                Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## Intercept          0.16      1.07    -1.94     2.25 1.00     2424     2399
## social.support     6.58      1.18     4.27     8.94 1.00     2513     2230
##
## Further Distributional Parameters:
##       Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## sigma     0.37      0.03     0.31     0.43 1.00     2619     1861
##
## Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
## and Tail_ESS are effective sample size measures, and Rhat is the potential
## scale reduction factor on split chains (at convergence, Rhat = 1).
```

```r
plot(m1)
```




```r
fixed <- data.frame(fixef(m1))
kable(fixed)
```

	Estimate	Est.Error	Q2.5	Q97.5
Intercept	0.1637425	1.073648	-1.938740	2.248606
social.support	6.5803637	1.183689	4.265843	8.941612

Part 3: Interpretation

Assignment 5.

Interpret the results against the background of the hypothesis.

Model `m1` seemed to have converged on stable parameter estimates: The Gelman-Rubin criterion `Rhat` is at 1, and the effective sample size (ESS) is above 700 for all parameters. Graphically the MCMC chains mixed well, showing a horizontal zig-zag without systematic drift upwards or downwards ("hairy caterpillar"). The posterior distributions are all unimodal with relatively smooth surfaces.

The posterior mean for the effect of social support on happiness was 6.58, 95% CrI[4.27, 8.94]. This means that according to the model, an increase in social support by one unit on the scale of social support was associated with an increase of around 6.5 units on the happiness scale, which is arguably substantial. There was thus positive evidence for the hypothesis that social support promotes happiness. Furthermore, there was also evidence that social support varies between countries. We could see this in the Bayes factor for the comparison of models `m1` and `m0`: $BF_{10} > 100$. We can also see this when computing the intraclass correlation for `m1` (ICC), which tells us the proportion of the variance in the data attributable to the grouping level 2 relative to the individual level 1. In this case the ICC is 0.26, i.e. it is above 10%, providing further suport for `m1` over `m0`.